

Piyush Rajendra Ladukar

Full Stack Developer | Python | Machine Learning | Building in GenAI & Agentic AI

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SUMMARY

Final-year B.Tech CSE student with hands-on experience in backend development, building secure REST APIs, and developing scalable, data-driven applications using real-world datasets. Passionate about AI and currently building LLM-powered applications while expanding expertise in Generative AI, Retrieval-Augmented Generation (RAG), and AI agents. Strong foundation in Python, backend systems, and problem-solving, with a proven track record of delivering end-to-end projects. Seeking Backend Developer, Python Developer, or AI/ML Engineer internships and full-time opportunities.

EXPERIENCE

[Smart Start Pvt. Ltd.](#) — Full Stack Developer Intern

Jul 2026 – Present | On-site

- Developed and maintained **10+** full-stack features using **Python, FastAPI, React.js, and Next.js** for scalable business applications.
- Built and integrated **15+ REST API endpoints**, enabling seamless communication between frontend interfaces and backend services.
- Designed and optimized **8+ database models** while implementing **JWT-based authentication** and role-based access workflows to improve application security.
- Collaborated with a cross-functional team using **Git** to develop, test, review, and deploy **20+ production-ready code changes** across multiple development sprints.
- Stack:** Python, FastAPI, Flask, React.js, Next.js, Node.js, Express.js, JavaScript, TypeScript, PostgreSQL, MongoDB, REST APIs, Git, Docker

[ESM Elixir Statergic Management Pvt. Ltd.](#) — Backend Developer Intern

Aug 2025 – Oct 2025 | On-site, Nagpur

- Built and maintained **12 REST API endpoints** using Node.js, Express.js, and PostgreSQL for internal enterprise applications supporting CIRP/IBC advisory workflows. [Verify: endpoint count]
- Implemented JWT Authentication and RBAC-based authorization systems, improving secure access control and backend security workflows.
- Collaborated on secure enterprise systems integrating authentication, role management, and database-driven services.
- Stack:** JavaScript (Node.js, Express.js), JWT, RBAC, REST APIs, SQL, MongoDB, Git, Postman

PROJECTS

[SecondBrain](#) — Personal AI Agent & RAG System (In Development)

- Building a Retrieval-Augmented Generation (RAG) pipeline with LangChain and a vector database to power a personal knowledge-agent system with LLM-based reasoning and tool calling.
- Designing agent memory and multi-step workflow logic for context retrieval, embeddings generation, and structured LLM responses.

[LokDrishti](#) — Parliamentary Intelligence Platform | [Github](#)

- Built a full-stack civic data platform scraping and structuring Lok Sabha MP data (attendance, debates, criminal records) for **543 MPs**; deployed Flask APIs and a React dashboard via Supabase with OAuth.

[AatankDrishti](#) — Global Terrorism Analytics | [Github](#)

- Engineered a data pipeline and React dashboard analyzing **181K+** terrorism incidents across **200+** countries from the GTD dataset.

[Why Farmer Suicides](#) — Agricultural Crisis Analytics | [Github](#)

- Analyzed Maharashtra farmer suicide data (2014–2023) across **33 districts**, building dashboards correlating debt, crop failure, and weather patterns.

[Anomax](#) — ML Anomaly Detection | Published, IJVRA

- Developed a machine learning model to detect anomalous electricity consumption patterns; published findings in the **International Journal of Versatile Research and Analysis (IJVRA)**.

OTHER PROJECTS & OPEN SOURCE

- [Blocks by Pi](#) — computer vision hand-gesture recognition system for virtual block placement (Python, OpenCV, MediaPipe).
- [QR-Scanner](#) — real-time QR detection/decoding app with clipboard integration (Python, OpenCV, Pyzbar).
- [Rajinoti](#) — contributor to an **open-source civic-tech platform** for structured **political data and analytics**.

SKILLS

Languages: Python, JavaScript, SQL

Backend: FastAPI, Flask, Node.js, Express.js, REST APIs, JWT, OAuth, RBAC

AI/ML: Machine Learning, Computer Vision, Pandas, NumPy, Scikit-learn, OpenCV

Gen AI : LangChain, RAG, Vector Databases, LLM APIs, Embeddings, Agentic AI

Databases: PostgreSQL, MySQL, MongoDB, Supabase, SQLAlchemy

Tools & Platforms: Git, GitHub, Docker, Linux, Vercel, Render, Postman

EDUCATION

Jhulelal Institute of Technology, Nagpur — B.Tech, Computer Science

CGPA: 8.4 | July 2023 – June 2027

ACHIEVEMENTS

- Serving as Secretary of Encypherist – CSE Technical Forum during the 2026–2027 tenure.
- Participated in 12+ hackathons; organized 2 hackathon events as Event Head.
- Achieved 2 Gold Medals at the National Karate Championship (2019) among 250+ athletes.
- Built and deployed multiple full-stack and AI-based applications, each live in production.